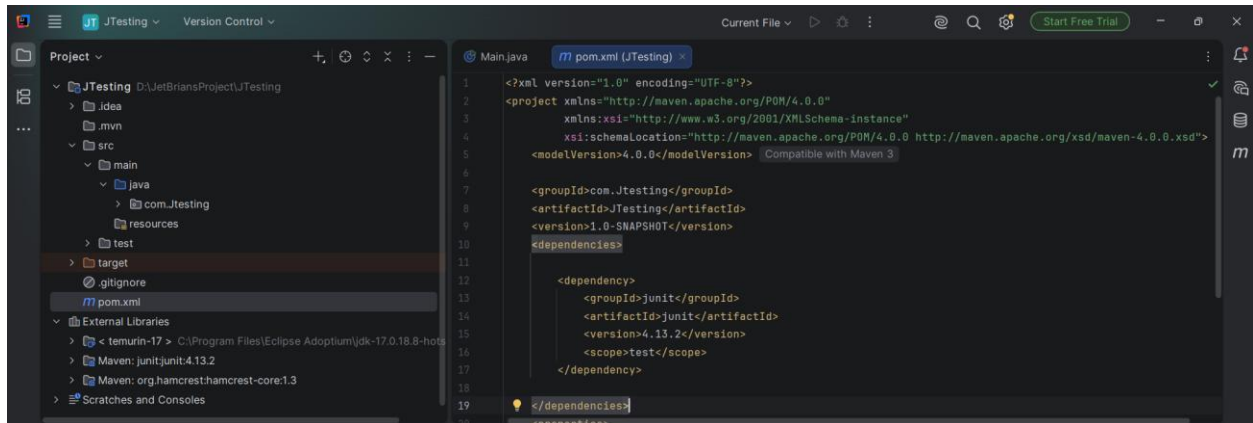


Exercise 1: Setting Up Junit



Exercise 2: Writing Basic JUnit Tests

Calculator.java

```
package com.Jtesting;

public class Calculator {

    public int add(int a, int b) {
        return a + b;
    }

    public int subtract(int a, int b) {
        return a - b;
    }

    public int multiply(int a, int b) {
        return a * b;
    }

    public int divide(int a, int b) {
        return a / b;
    }
}
```

CalculatorTest.java

```
package com.Jtesting;

import org.junit.Test;

import static org.junit.Assert.*;

public class CalculatorTest {

    Calculator calculator = new Calculator();

    @Test
```

```

public void testAddition() {
    assertEquals(10, calculator.add(5, 5));
}

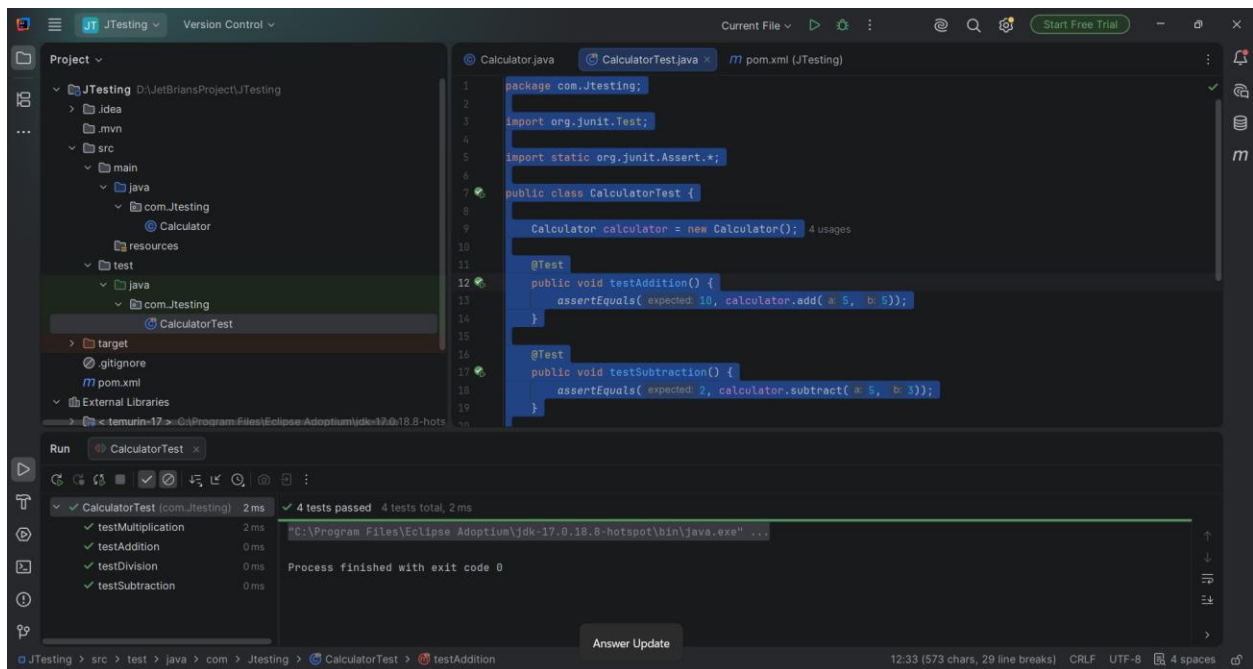
@Test
public void testSubtraction() {
    assertEquals(2, calculator.subtract(5, 3));
}

@Test
public void testMultiplication() {
    assertEquals(20, calculator.multiply(4, 5));
}

@Test
public void testDivision() {
    assertEquals(5, calculator.divide(20, 4));
}
}

```

Output:



Exercise 3: Assertions in Junit

AssertionTest.java:

```
package com.Jtesting;

import org.junit.Test;
import static org.junit.Assert.*;

public class AssertionsTest {

    @Test
    public void testAssertions() {

        assertEquals(5, 2 + 3);

        assertTrue(5 > 3);

        assertFalse(5 < 3);

        assertNull(null);

        assertNotNull(new Object());

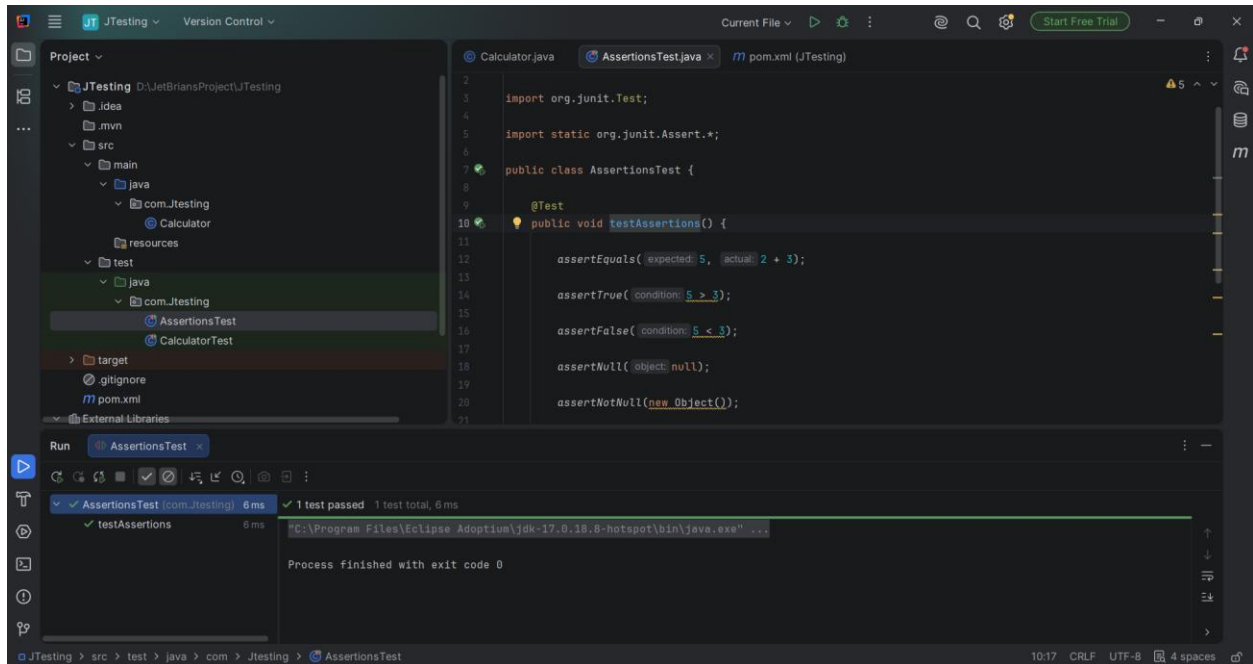
        String s1 = "Java";
        String s2 = s1;

        assertEquals(s1, s2);

        String s3 = new String("Java");

        assertNotSame(s1, s3);
    }
}
```

Output



Exercise 4: Arrange-Act-Assert (AAA) Pattern, Test Fixtures, Setup and Teardown Methods in Junit

CalculatorAAATest.java

```
package com.Jtesting;

import org.junit.After;
import org.junit.Before;
import org.junit.Test;

import static org.junit.Assert.*;

public class CalculatorAAATest {

    private Calculator calculator;

    @Before
    public void setUp() {
        System.out.println("Setup");
        calculator = new Calculator();
    }

    @After
    public void tearDown() {
        System.out.println("Teardown");
    }

    @Test
    public void testAdd() {
```

```
// Arrange
int a = 15;
int b = 25;

// Act
int result = calculator.add(a, b);

// Assert
assertEquals(40, result);
}
```

Output:

